

7-Days AWS DevOps Series – Beginner to Advanced

17 **Day 1: AWS Foundations**

Goal: Build core understanding needed for DevOps

Topics:

- What is AWS? Global infrastructure (regions, AZs)
 - **IAM Basics** → Users, Groups, Policies
 - **IAM Roles** → EC2 Role, Lambda Role, Service-linked roles
 - **AWS CLI setup**
 - **S3 Introduction** → Buckets, versioning, static hosting
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17 **Day 2: Networking & Compute**

Goal: Learn core networking + compute layers a DevOps engineer must master

Topics:

- **VPC, Subnets (Private/Public)**
 - **Internet Gateway, NAT Gateway, Route Tables**
 - **Security Groups vs NACL**
 - **EC2 Instances setup**
 - Key pairs, SSH, Elastic IP, Auto-recovery
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Day 3: DevOps CI/CD Foundations in AWS

Goal: Understand AWS-native DevOps ecosystem

Topics:

- Introduction to AWS DevOps tools
 - **CodeCommit** (Git repo)
 - **AWS Artifact** (Compliance reports if needed for enterprises)
 - CI/CD concepts: Pipeline, Stages, Build, Deploy
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Day 4: Build Stage – AWS CodeBuild

Goal: Learn Build automation

Topics:

- Creating **buildspec.yml**
 - CodeBuild compute environment
 - Artifacts storage
 - Environment variables & secrets
 - Integrate CodeBuild + S3 + CloudWatch Logs
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Day 5: Deploy Stage – AWS CodeDeploy

Goal: Master blue-green & rolling deployments

Topics:

- AppSpec file for EC2 & ECS
- Deployment groups

- Hooks (BeforeInstall, AfterInstall)
 - Blue/Green deployment with Load Balancer
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Day 6: CI/CD End-to-End Automation – CodePipeline

Goal: Deploy fully automated pipeline

Topics:

- Source → Build → Deploy
 - Integrating CodeCommit → CodeBuild → CodeDeploy
 - Manual approvals
 - Notifications using SNS
 - Artifact handling
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Day 7: Real Project Implementation

Goal: Build & deploy a real application

Project:

- ✓ Deploy a sample application using **CodeCommit** → **CodeBuild** → **CodeDeploy** → **EC2**
 - ✓ S3 for logs + IAM roles + VPC networking
 - ✓ Output: Fully automated CI/CD pipeline
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3-Month AWS DevOps Learning Timeline



Month 1 — AWS Core + Networking + Compute

Week 1–2: Fundamentals

- IAM, S3, EC2
- AWS Shared Responsibility Model
- Security Groups, NACL

Week 3: VPC & Networking

- Custom VPC creation
- Subnets, routing, NAT, IGW
- VPC Peering & Endpoints

Week 4: Load Balancing + Scaling

- ALB, NLB
 - Auto Scaling Group
 - Launch templates
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Month 2 — DevOps Services Deep Dive

Week 1: CodeCommit

- Repo creation
- Triggers
- Branch policies

Week 2: CodeBuild

- Buildspec.yml
- Build environments
- Storing artifacts

Week 3: CodeDeploy

- EC2 / ECS deployment
- Rolling vs Blue-Green strategies

Week 4: CodePipeline

- Multi-stage pipelines
 - Manual approvals
 - SNS notifications
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Month 3 — Automation + Monitoring + Projects

Week 1: Infrastructure as Code

- **CloudFormation** basics
- Template structure
- EC2 stack creation

Week 2: Monitoring & Logging

- CloudWatch Logs
- CloudWatch Alarms
- X-Ray (optional)

Week 3: Container DevOps (Bonus)

- ECR, ECS, Fargate
- CI/CD for containers

Week 4: Final Project + Certification Prep

- Build final project
 - Prepare for **AWS Certified Developer / DevOps Engineer** exam
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Important AWS Services for DevOps (Must Learn)

IAM

- Users, Groups, Policies
- Roles (EC2 Role, CodeBuild Role, CodeDeploy Role)
- Service-linked roles

S3

- Store logs, artifacts, versioning
- Backend for Terraform (optional)

EC2

- Application servers
- Deployment target for CodeDeploy
- Auto Scaling

VPC

- Private & Public subnets
- SG vs NACL
- Load Balancers

AWS DevOps Tools

CodeCommit

- Git repository system
- Ideal for CI/CD projects

CodeBuild

- Builds your application
- Executes buildspec.yml
- Generates artifacts

CodeDeploy

- Automates deployment to EC2/ECS
- Blue/Green deployments

CodePipeline

- Orchestrates the CI/CD workflow
- Connects Source → Build → Deploy

AWS Artifact (Compliance)

- Provides compliance & security reports for enterprises
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Project Ideas for AWS DevOps



Project 1: End-to-End CI/CD Pipeline for EC2

- CodeCommit → CodeBuild → CodeDeploy → EC2
- S3 for artifacts
- Blue/Green deployment with ALB



Project 2: Container CI/CD Pipeline

- Build Docker image with CodeBuild
- Push to ECR
- Deploy automatically to ECS Fargate via CodePipeline



Project 3: Serverless CI/CD

- Lambda function deployment using CodePipeline
- SAM/CloudFormation templates
- S3 → CodeBuild → Lambda deployment



Project 4: Multi-Environment Pipeline

- Dev → Staging → Prod
 - Manual approvals in CodePipeline
 - Cross-account roles (optional advanced topic)
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Best AWS Certifications for DevOps Engineers

- ✓ AWS Certified Developer – Associate
 - ✓ AWS Certified Solutions Architect – Associate
 - ✓ AWS DevOps Engineer – Professional
 - ✓ AWS SysOps Administrator – Associate (optional)
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Best 3 YouTube Channels for Learning AWS

- 1 [Stephane Maarek – AWS Certification & Hands-on](#)
- 2 [Abhishek Veeramalla](#)
- 3 [FreeCodeCamp AWS Playlists](#)